

Large MDPs

* So far, we have been focusing on MDPs with finite (discrete) state and action spaces.

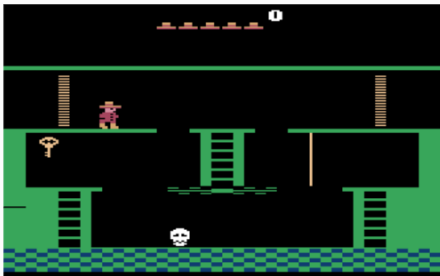
Example:
tabular Q

s	a	$Q(s,a)$

* Also, we have been working in the tabular setting, i.e., algorithms that use a tabular representation for the value function(s) and policy.

However, the tabular setting does not scale well to

- Large finite (discrete) state and action spaces,
- infinite discrete state and action spaces, and
- continuous state and action spaces.



Idea: We will use function approximation techniques to generalize the methods learned in the tabular setting.

→ Supervised Learning



Supervised Learning

A branch of machine learning where given

a labeled data set $\{(x_i, y_i)\}_{i=1}^N$, $\begin{cases} x_i \in X & \text{feature} \\ y_i \in Y & \text{label} \end{cases}$,

we are interested in learning a function f that predicts labels for input features, i.e.,

$$f: X \rightarrow Y.$$

Regression: Consider that there is a data distribution D , i.e., a joint probability distribution, over the features x and label y :

Example:

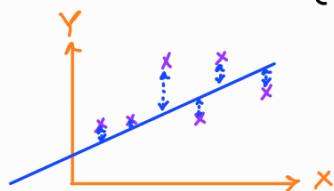
$P_Y \in \Delta(\{\text{Cat}, \text{Not Cat}\})$
 $Y \sim P_Y$
 $X|Y \sim P_{X|Y}$

$(x, y) \stackrel{\text{i.i.d.}}{\sim} D.$

Example:

$X \sim \text{Unif}(-B, B)$
 $Y|X \sim \mathcal{N}(x, \sigma^2)$

suppose that we look to learn a function restricted to a class of functions $\mathcal{F} = \{f: X \rightarrow \mathbb{R}\}.$



Examples:

- $\mathcal{F} = \{f_\theta: f_\theta = \sum_{i=1}^n \theta_i x_i\}$
- $\mathcal{F} = \{f_\theta: f_\theta = \mathcal{NN}(\theta; \cdot)_{i=1}^n\}$

We need an error metric (loss function) $L(\hat{y}, y)$ to capture the difference between the predicted labels and the true labels,

for instance, $\|\hat{y} - y\|_2$, $\|\hat{y} - y\|_2^2$, or $\|\hat{y} - y\|_1$.

The objective is to minimize the error over the entire data distribution:

$$\min_{\substack{f \in \mathcal{F} \\ \theta \in \Theta}} \mathbb{E}_{(x,y) \sim \mathcal{D}} [L(\hat{y}, y)]$$

$\downarrow \hat{y} = f(x)$

x_1
 x_2
 $\vdots$
 x_N

f_θ

$\hat{y}_1$
 $\hat{y}_2$
 $\vdots$
 $\hat{y}_N$

while the optimal mapping is

$$f^* = \arg \min_{f \in \mathcal{F}} \mathbb{E}_{(x,y) \sim \mathcal{D}} [L(\hat{y}, y)]$$

However, we do not have access to the distribution, rather, data sampled from the distribution. $\longrightarrow$

In this case, we can do empirical risk minimization (ERM):

$$\mathbb{E}_{(x,y) \sim \mathcal{D}} [L(f(x), y)] \approx \overbrace{\hat{\mathbb{E}} [L(f(x), y)]}^{\text{empirical mean}} = \frac{1}{N} \sum_{i=1}^N L(f(x_i), y_i)$$

$$\hat{f} = \arg \min_{f \in \mathcal{F}} \sum_{i=1}^N L(f(x_i), y_i)$$

* **Result:** Let $L(\hat{y}, y) = \|\hat{y} - y\|_2^2$. Under the assumption of realizability, i.e., $f^* \in \mathcal{F}$, it holds that:

$$\mathbb{E}_{(x,y) \sim \mathcal{D}} [(\hat{f}(x) - f^*(x))^2] \leq \delta$$

$\delta \downarrow$ as $N \uparrow$
 e.g., $\delta = \frac{c}{\sqrt{N}}$